

Tutorial 12

Diagonalization and applications

1. For each of the following statements, say whether it is true or false. Give a reason for your answer,
 - (a) Every diagonalizable matrix is invertible.
 - (b) Every invertible matrix is diagonalizable.
 - (c) If a square matrix has a repeated eigenvalue, it is not diagonalizable.
2. In Tutorial 11, you found eigenvalues and bases for the corresponding eigenspaces for three matrices, in Questions 3, 5 and 6. Use your answers to those questions to determine whether the matrices are diagonalizable; give a reason in each case. In the case(s) where the matrix is diagonalizable, give an invertible matrix P which diagonalizes the matrix, and the corresponding diagonal matrix D .
 - (a) The matrix A in Question 3.
 - (b) The matrix A in Question 5.
 - (c) The matrix B in Question 6.
3. Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & -3 & -1 \end{pmatrix}$.
 - (a) Find the eigenvalues of A .
 - (b) Use only your answer to (a) to prove or disprove the following statements:
 - i. A is diagonalisable.
 - ii. A is invertible.
 - (c) If A is diagonalisable, find an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$.
 - (d) Prove or disprove:
 - i. There is a $k \in \mathbb{N}$ such that A^k is the zero matrix.
 - ii. There is a $k \in \mathbb{N}$ such that $A^k = I$.
4. In this question we use diagonalization to solve a system of linear differential equations. You will need to know that the general solution of the differential equation $x'(t) = \frac{dx}{dt} = kx$ is $x(t) = Ce^{kt}$, with C a constant. Consider the system of differential equations:

$$\begin{aligned}x_1'(t) &= x_1(t) + 3x_2(t) \\ x_2'(t) &= 4x_1(t) + 5x_2(t).\end{aligned}$$

- (a) Write $\mathbf{x}(t)$ for $\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$ and $\mathbf{x}'(t)$ for $\begin{pmatrix} x_1'(t) \\ x_2'(t) \end{pmatrix}$.

Use this notation to write system above in the form

$$\mathbf{x}'(t) = A\mathbf{x}(t), \quad (*)$$

for some matrix A .

- (b) Find an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$.
- (c) Let $\mathbf{x}(t) = P\mathbf{u}(t)$, where $\mathbf{u}(t) = \begin{pmatrix} u_1(t) \\ u_2(t) \end{pmatrix}$. Show that $\mathbf{x}'(t) = P\mathbf{u}'(t)$, where $\mathbf{u}'(t) = \begin{pmatrix} u_1'(t) \\ u_2'(t) \end{pmatrix}$. Substitute in equation (*).
- (d) You should now have two simple differential equations for u_1 and u_2 . Solve them.
- (e) Use your solutions for u_1 and u_2 to find solutions for x_1 and x_2 .
5. Excessive university fees have driven you to selling subscriptions to a pathetic magazine entitled “How to Succeed in Linear Algebra” to make ends meet. Every month, you send an email to a large mailing list. Some of the students receiving this email already subscribe to your magazine, others do not. You notice the following: from the mailing list, 60 percent of those who already subscribe will subscribe again, while 25 percent of those who do not now subscribe will subscribe.
- (a) Write down the transition matrix M for this process: that means the matrix M satisfying $\mathbf{u} = M\mathbf{x}$ where

$$\mathbf{u} = \begin{pmatrix} \text{fraction of people on mailing list subscribing after letter} \\ \text{fraction of people on mailing list not subscribing after letter} \end{pmatrix}$$

$$\mathbf{x} = \begin{pmatrix} \text{fraction of people on mailing list subscribing before letter} \\ \text{fraction of people on mailing list not subscribing before letter} \end{pmatrix}$$

- (b) M should be a Markov matrix: is it?
- (c) Find an invertible matrix P and a diagonal matrix D such that $P^{-1}MP = D$. [Shortcut: a Markov matrix always has 1 as eigenvalue; what is the other eigenvalue?].
- (d) Suppose you begin with nobody subscribing. According to this model, what fraction of the people on the mailing list will subscribe after your first letter?
- (e) Use $M = PDP^{-1}$ to find an expression for M^n in terms of P , P^{-1} and D^n . Find $A = \lim_{n \rightarrow \infty} M^n$. What does this mean for your subscription situation in the long run?
- (f) The columns of A should all be the same. Let $\mathbf{v}$ be one of these columns. What is $M\mathbf{v}$?

6. Let A be an $n \times n$ real matrix. We call a subspace S of $\mathbb{R}^n$ **invariant** under the matrix A if $A\mathbf{x} \in S$ for every $\mathbf{x} \in S$ (i.e. A sends every vector in S to a (possibly different) vector in S , never to a vector outside S). The trivial subspace $\{\mathbf{0}\}$ and $\mathbb{R}^n$ itself are always invariant subspaces. Show a non-trivial proper subspace of $\mathbb{R}^2$ is an invariant subspace of a 2×2 matrix A if and only if it is an eigenspace of A .